Lab Exercise 8: Public-Private Key Generation

Core Exercise:

This week aims to look at RSA and expand on the work you put into implementing Diffie-Hellman last over the last two weeks. This exercise should be considered only if you are happy with the DH key passing that we have already discussed and you have implemented some form of encryption on your current testbed/program. Remember that cryptography is hard – we are looking at a field where we get very little feedback and it is difficult to tell when something has gone wrong in an algorithm.

> *"There are known knowns. These are things we know that we know. There are known unknowns. That is to say, there are things that we know we don't know. But there are also unknown unknowns. There are things we don't know we don't know."*

Donald Rumsfield

Read through the following articles:

http://www.muppetlabs.com/~breadbox/txt/rsa.html

https://en.wikibooks.org/wiki/Cryptography/A_Basic_Public_Key_Example

Create a simple implementation of RSA and encrypt a variety of texts using your key pairs, examples of this are here in C, Python, and Java in case you get stuck

C: https://shanetully.com/2012/04/simple-public-key-encryption-with-rsa-and-openssl/

Python: https://gist.github.com/tylerl/1239116

Java: http://pajhome.org.uk/crypt/rsa/implementation.html

The following resources may be useful in creating your implementation. https://www.di-mgt.com.au/rsa_alg.html#simpleexample

https://en.wikipedia.org/wiki/RSA_(cryptosystem)

Exercise 2:

Create a simple server which a user can log into a provide a username and a public key pair which can then be requested by other users. The server should provide a unique ID for each registered user and store the details to disk when the server is shut down. The server should keep a log of all requests and store this on disk. You should implement the following commands:

Client:

REGISTER <username> <key>

GET <username>

Server:

SUCCESS REGISTER <username>

SUCCESS GET <username> <key>

FAIL REGISTER inuse

FAIL REGISTER tooshort

FAIL GET nosuchid

ERROR unknown